TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	81.7 ± 1.27
Correlation Stability	87.1 ± 0.54
Distribution Consistency	87.1 ± 0.83
Entropy Preservation	81.9 ± 1.37
Mean Z Score Drift	0.4 ± 0.02
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	19.4 ± 2.76
Data Density	100.0 ± 0.00
Coverage Score	41.2 ± 4.58
Feature Accuracy	67.8 ± 1.94
Novelty Score	0.5 ± 0.02

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Dataset Audit: indian_diabetes.csv

1. Dataset Portrait

Attribute	Value
Rows	768
Columns	9
Numeric features	9
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	google/gemma-4-e2b
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Pregnancies	3.845	4.403	+0.558	14.5%	Warn
Glucose	120.895	139.161	+18.267	15.1%	Warn
BloodPressure	69.105	67.769	-1.337	1.9%	OK
SkinThickness	20.536	19.855	-0.682	3.3%	OK
Insulin	79.799	78.812	-0.988	1.2%	OK
BMI	31.993	35.490	+3.498	10.9%	Warn
DiabetesPedigreeFunction	0.472	0.573	+0.101	21.5%	Warn
Age	33.241	35.645	+2.404	7.2%	OK
Outcome	0.349	1.000	+0.651	186.6%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	81.7 ± 1.27
Correlation Stability (*2)	87.1 ± 0.54
Distribution Consistency (*3)	87.1 ± 0.83
Entropy Preservation (*4)	81.9 ± 1.37
Mean Z Score Drift (*5)	0.4 ± 0.02
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	19.4 ± 2.76
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	41.2 ± 4.58
Feature Accuracy (*10)	67.8 ± 1.94

Novelty Score (*11)	0.5 ± 0.02

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.772077922077922	N/A	0.7311688311688311	-0.04090909090909096	*
F1	0.6262914445612621	N/A	0.6453853356188921	0.01909389105762975	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7103896103896103	N/A	0.6824675324675324	-0.02792207792207792	
F1	0.5687460273652091	N/A	0.5725830101063426	0.00383698274113364 32	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7292207792207792	N/A	0.7207792207792209	-0.00844155844155844	
F1	0.592152063889281	N/A	0.6196230637615668	0.02747099987228583	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.746103896103896	N/A	0.7383116883116883	-0.00779220779220786	
F1	0.6156864582943115	N/A	0.6484228081002363	0.03273634980592505	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7493506493506492	N/A	0.7616883116883117	0.012337662337662359	
F1	0.6076604413342895	N/A	0.6781541181185525	0.07049367678426294	*

Model: KNN

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.754545454545454 5	N/A	0.691558441558441 5	-0.06298701298701301	*
F1	0.599825946338941 6	N/A	0.620101352675435 6	0.02027540633649399 5	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.763636363636363 6	N/A	0.753246753246753 2	-0.01038961038961041 6	
F1	0.639985235463502 5	N/A	0.655190227346630 3	0.01520499188312790 5	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.685064935064935	N/A	0.687662337662337 6	0.00259740259740257 6	
F1	0.462363420876162 06	N/A	0.569062208734137 2	0.1066987878579753	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.756493506493506 4	N/A	0.734415584415584 4	-0.02207792207792210 6	
F1	0.573368073405794 2	N/A	0.634442211680535 6	0.06107413827474141	*

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.754545454545454 5	N/A	0.753896103896103 8	-0.00064935064935066 63	
F1	0.612681724368225 4	N/A	0.667262054008902 9	0.05458032964067748	*

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunct	Age	Outcome
1	6.0	148.0	72.0	35.0	0.0	33.6	0.627	50.0	1.0
2	1.0	85.0	66.0	29.0	0.0	26.6	0.351	31.0	0.0
3	8.0	183.0	64.0	0.0	0.0	23.3	0.672	32.0	1.0
4	1.0	89.0	66.0	23.0	94.0	28.1	0.167	21.0	0.0
5	0.0	137.0	40.0	35.0	168.0	43.1	2.288	33.0	1.0
6	5.0	116.0	74.0	0.0	0.0	25.6	0.201	30.0	0.0
7	3.0	78.0	50.0	32.0	88.0	31.0	0.248	26.0	1.0
8	10.0	115.0	0.0	0.0	0.0	35.3	0.134	29.0	0.0
9	2.0	197.0	70.0	45.0	543.0	30.5	0.158	53.0	1.0
10	8.0	125.0	96.0	0.0	0.0	0.0	0.232	54.0	1.0
11	4.0	110.0	92.0	0.0	0.0	37.6	0.191	30.0	0.0
12	10.0	168.0	74.0	0.0	0.0	38.0	0.537	34.0	1.0
13	10.0	139.0	80.0	0.0	0.0	27.1	1.441	57.0	0.0
14	1.0	189.0	60.0	23.0	846.0	30.1	0.398	59.0	1.0
15	5.0	166.0	72.0	19.0	175.0	25.8	0.587	51.0	1.0
16	7.0	100.0	0.0	0.0	0.0	30.0	0.484	32.0	1.0
17	0.0	118.0	84.0	47.0	230.0	45.8	0.551	31.0	1.0
18	7.0	107.0	74.0	0.0	0.0	29.6	0.254	31.0	1.0
19	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0
20	1.0	115.0	70.0	30.0	96.0	34.6	0.529	32.0	1.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	2.0	114.0	79.0	41.0	187.0	37.655634545418614	0.9223792210208072	29.0	1.0	1.0
2	12.0	129.0	0.0	29.0	0.0	39.962810834002475	0.5598547425692394	43.0	1.0	1.0
3	10.0	116.0	67.0	29.0	0.0	32.0327627607998	0.24025757870364514	39.0	1.0	1.0
4	8.0	129.0	70.0	0.0	0.0	34.14194565548066	0.5749328576505792	46.0	1.0	1.0
5	4.0	171.0	86.0	27.0	155.0	33.52441371496956	1.161854480023934 5	49.0	1.0	1.0
6	7.0	175.0	79.0	40.0	313.0	37.98999101909646	0.5631304263568191	40.0	1.0	1.0
7	8.0	110.0	76.0	0.0	0.0	27.8	0.237	58.0	0.0	nan
8	1.0	90.0	62.0	18.0	59.0	25.1	1.268	25.0	0.0	nan
9	2.0	56.0	56.0	28.0	45.0	24.2	0.332	22.0	0.0	nan
10	5.0	111.0	72.0	28.0	0.0	23.9	0.407	27.0	0.0	nan

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	3.0	96.0	67.0	32.0	162.0	37.307784354204976	0.5985849270107525	25.0	1.0	2.0
2	0.0	120.0	66.0	30.0	165.0	34.29974587572796	0.2031448508350607	33.0	1.0	2.0
3	5.0	117.0	76.0	0.0	0.0	31.895640826955688	0.2836029755445527 7	46.0	1.0	2.0
4	4.0	178.0	71.0	33.0	138.0	30.832846686246214	0.1881685952079102	33.0	1.0	2.0
5	4.0	113.0	69.0	0.0	0.0	32.628516130475099	0.3278225443815 0166	43.0	1.0	2.0
6	11.0	107.0	69.0	0.0	0.0	29.987515165701492	0.5501930133572899	36.0	1.0	2.0
7	0.0	91.0	68.0	32.0	210.0	39.9	0.381	25.0	0.0	nan
8	3.0	173.0	82.0	48.0	465.0	38.4	2.137	25.0	1.0	nan
9	1.0	147.0	94.0	41.0	0.0	49.3	0.358	27.0	1.0	nan
10	2.0	107.0	74.0	30.0	100.0	33.6	0.404	23.0	0.0	nan

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B.3 Audit Trace: Fold 3 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	4.0	129.0	64.0	29.0	114.0	27.652242159067228	0.21127588948612736	29.0	1.0	3.0
2	5.0	129.0	74.0	0.0	0.0	28.999647006863785	0.19588550669292493	56.0	1.0	3.0
3	5.0	119.0	92.0	0.0	0.0	48.328936182023696	0.4564336718861084	30.0	1.0	3.0
4	7.0	132.0	74.0	0.0	0.0	34.48483730093056	0.44535457878038087	38.0	1.0	3.0
5	3.0	147.0	61.0	28.0	322.0	30.576780553091933	0.17084208157648584	29.0	1.0	3.0
6	11.0	148.0	86.0	30.0	147.0	35.136809315570304	0.7412868370267383	51.0	1.0	3.0
7	5.0	158.0	84.0	41.0	210.0	39.4	0.395	29.0	1.0	nan
8	1.0	101.0	50.0	15.0	36.0	24.2	0.526	26.0	0.0	nan
9	0.0	109.0	88.0	30.0	0.0	32.5	0.855	38.0	1.0	nan
10	5.0	147.0	75.0	0.0	0.0	29.9	0.434	28.0	0.0	nan

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	14.0	135.0	70.0	31.0	110.0	36.913275201493896	0.17556637503896944	43.0	1.0	4.0
2	5.0	98.0	76.0	26.0	0.0	35.101947915865814	0.4388431194672032	51.0	1.0	4.0
3	6.0	142.0	81.0	38.0	357.0	44.212806092347504	0.3743711819126502	44.0	1.0	4.0
4	9.0	102.0	76.0	36.0	0.0	32.563503034284494	0.7091741761375461	46.0	1.0	4.0
5	5.0	138.0	108.0	0.0	0.0	48.291063527089004	0.22934893756728136	37.0	1.0	4.0
6	10.0	148.0	78.0	44.0	126.0	38.531716848868896	0.6749030109321304	41.0	1.0	4.0
7	5.0	117.0	86.0	30.0	105.0	39.1	0.251	42.0	0.0	nan
8	3.0	74.0	68.0	28.0	45.0	29.7	0.293	23.0	0.0	nan
9	2.0	129.0	0.0	0.0	0.0	38.5	0.304	41.0	0.0	nan
10	1.0	126.0	56.0	29.0	152.0	28.7	0.801	21.0	0.0	nan

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B.5 Audit Trace: Fold 5 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	1.0	167.0	73.0	17.0	143.0	23.6897215690600 3	0.43974232837102 145	32.0	1.0	5.0
2	1.0	168.0	88.0	29.0	89.0	36.1016822603868 3	0.30656473552722 086	35.0	1.0	5.0
3	4.0	114.0	85.0	0.0	0.0	17.1051857068649 1	0.24910518570686 49	52.0	1.0	5.0
4	12.0	136.0	73.0	34.0	104.0	36.5914793530548 86	0.18520630763985 724	41.0	1.0	5.0
5	1.0	132.0	42.0	31.0	169.0	39.2689598856887 1	2.05574319306987 8	33.0	1.0	5.0
6	4.0	167.0	75.0	40.0	474.0	36.9493792551866 5	1.61147966653807 26	31.0	1.0	5.0
7	13.0	126.0	90.0	0.0	0.0	43.4	0.583	42.0	1.0	nan
8	1.0	119.0	44.0	47.0	63.0	35.5	0.28	25.0	0.0	nan
9	1.0	181.0	78.0	42.0	293.0	40.0	1.258	22.0	1.0	nan
10	3.0	78.0	70.0	0.0	0.0	32.5	0.27	39.0	0.0	nan

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	1.0	125.0	93.0	39.0	64.0	32.4629150161731 4	1.17998896430418 25	32.0	1.0	6.0
2	7.0	169.0	88.0	32.0	297.0	33.3781001212891 8	0.46415970695255 154	58.0	1.0	6.0
3	1.0	141.0	0.0	0.0	0.0	41.3245294876409 94	0.54653501521648 2	28.0	1.0	6.0
4	3.0	122.0	66.0	32.0	252.0	38.1936401503542 3	0.52171210038444 07	29.0	1.0	6.0
5	2.0	125.0	59.0	0.0	0.0	30.8281385873645 1	0.52307063104182 8	43.0	1.0	6.0
6	6.0	179.0	84.0	36.0	289.0	35.1809221835083 86	0.37590084749933 26	45.0	1.0	6.0
7	0.0	102.0	52.0	0.0	0.0	25.1	0.078	21.0	0.0	nan
8	4.0	76.0	62.0	0.0	0.0	34.0	0.391	25.0	0.0	nan
9	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0	nan
10	1.0	124.0	74.0	36.0	0.0	27.8	0.1	30.0	0.0	nan

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B.7 Audit Trace: Fold 7 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	9.0	164.0	91.0	0.0	0.0	32.055544468607856	0.2957395377237518	48.0	1.0	7.0
2	4.0	176.0	66.0	35.0	519.0	32.23539611712155	0.34889357288337075	49.0	1.0	7.0
3	8.0	108.0	69.0	0.0	0.0	30.583810680596926	0.9248722658696316	33.0	1.0	7.0
4	3.0	178.0	74.0	32.0	190.0	34.7830412269728	0.7575118578620329	32.0	1.0	7.0
5	1.0	127.0	87.0	39.0	110.0	36.3570394174329	1.0500680740778512	36.0	1.0	7.0
6	7.0	134.0	75.0	27.0	130.0	32.7461872881902	0.4365950444041743	33.0	1.0	7.0
7	2.0	85.0	65.0	0.0	0.0	39.6	0.93	27.0	0.0	nan
8	2.0	142.0	82.0	18.0	64.0	24.7	0.761	21.0	0.0	nan
9	9.0	102.0	76.0	37.0	0.0	32.9	0.665	46.0	1.0	nan
10	4.0	117.0	64.0	27.0	120.0	33.2	0.23	24.0	0.0	nan

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	1.0	120.0	62.0	28.0	167.0	32.618634143040985	0.4633781847929581	33.0	1.0	8.0
2	5.0	147.0	84.0	37.0	0.0	43.984281518046544	0.4505139922246871	36.0	1.0	8.0
3	6.0	191.0	85.0	0.0	0.0	29.923093952924901	0.2087534165821096	62.0	1.0	8.0
4	10.0	105.0	70.0	27.0	0.0	27.926731586198418	0.1712591488395241	41.0	1.0	8.0
5	6.0	190.0	91.0	0.0	0.0	35.276978663746301	0.2752308184081834	65.0	1.0	8.0
6	8.0	125.0	70.0	23.0	117.0	30.468196486164806	0.4746652270414375	42.0	1.0	8.0
7	1.0	119.0	44.0	47.0	63.0	35.5	0.28	25.0	0.0	nan
8	8.0	196.0	76.0	29.0	280.0	37.5	0.605	57.0	1.0	nan
9	1.0	118.0	58.0	36.0	94.0	33.3	0.261	23.0	0.0	nan
10	1.0	116.0	70.0	28.0	0.0	27.4	0.204	21.0	0.0	nan

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B.9 Audit Trace: Fold 9 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	0.0	177.0	62.0	30.0	480.0	35.41417344586781	0.9924543185071608	21.0	1.0	9.0
2	10.0	100.0	75.0	32.0	200.0	38.046722522714704	0.5406549671985609	44.0	1.0	9.0
3	9.0	110.0	68.0	26.0	0.0	27.321708849045436	0.16863512839795758	38.0	1.0	9.0
4	5.0	108.0	80.0	39.0	183.0	33.919975137741304	0.512582425757345	32.0	1.0	9.0
5	8.0	181.0	77.0	32.0	0.0	34.96300464020838	0.3145840334472623	40.0	1.0	9.0
6	3.0	174.0	71.0	33.0	136.0	32.063093422949311	0.1935696784422165	28.0	1.0	9.0
7	8.0	65.0	72.0	23.0	0.0	32.0	0.6	42.0	0.0	nan
8	10.0	75.0	82.0	0.0	0.0	33.3	0.263	38.0	0.0	nan
9	2.0	146.0	70.0	38.0	360.0	28.0	0.337	29.0	1.0	nan
10	5.0	126.0	78.0	27.0	22.0	29.6	0.439	40.0	0.0	nan

B.10 Audit Trace: Fold 10 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	8.0	104.0	99.0	35.0	0.0	43.161479178886258	0.2486634763300758	45.0	1.0	10.0
2	6.0	193.0	77.0	0.0	0.0	23.492571944251228	0.14916717135791752	57.0	1.0	10.0
3	1.0	122.0	93.0	25.0	143.0	33.871068551791201	0.3404768148545381	41.0	1.0	10.0
4	6.0	149.0	79.0	38.0	174.0	38.206872745713109	0.7051147065455496	37.0	1.0	10.0
5	10.0	116.0	67.0	29.0	0.0	31.829850726886105	0.2358142494938573	39.0	1.0	10.0
6	2.0	171.0	76.0	20.0	118.0	31.120399042285793	0.2765058115068404	31.0	1.0	10.0
7	3.0	180.0	64.0	25.0	70.0	34.0	0.271	26.0	0.0	nan
8	3.0	99.0	80.0	11.0	64.0	19.3	0.284	30.0	0.0	nan
9	4.0	148.0	60.0	27.0	318.0	30.9	0.15	29.0	1.0	nan
10	7.0	136.0	90.0	0.0	0.0	29.9	0.21	50.0	0.0	nan